Intern GitHub Features overview

GitHub / Source Control training module. This is intentionally **lightweight, practical, and hands-on**, designed for interns who may be **new to Git entirely**.

🧠 Source Control Basics – GitHub Training for IT Interns

Goal:
Teach interns *why* source control matters, *how* to use GitHub safely, and *basic daily workflows* they'll encounter in IT, cloud, and automation work.

Total Time: ~2–3 hours
Skill Level: Beginner

📌 Section 1 — What Is Source Control? (15–20 min)

Key Concepts

- What is source control?
- Why IT teams use Git (not just developers)
- Real IT use cases:
  - Infrastructure as Code (Terraform, Bicep, CloudFormation)
  - Scripts (PowerShell, Bash, Python, etc.)
  - Documentation (Markdown: README.md)
  - Config backups (S3, Azure Blob)

Explain These Terms (Plain English)

Term	Meaning
Repository (Repo)	A project folder with history
Commit	A saved change
Branch	A copy to work safely
Pull Request (PR)	Request to merge changes
Merge	Combine changes

Term	Meaning
Version history	Who changed what and when

✅ **Checkpoint Question**

Why is Git better than saving scripts on a file share?

🧪 Lab 1 — GitHub Account & Repo Basics (20 min)

Objective

Learn how to navigate GitHub and create a repo.

Lab Steps

1. Log into GitHub
2. Create a new repository:
   - Name: `intern-training`
   - Initialize with README
3. Edit README in browser
4. Commit changes

✅ **Success Criteria**

- Repo exists
- Commit history shows changes
- Intern can explain what a commit is

📌 Section 2 — GitHub Workflow (Concepts) (15 min)

Explain the Standard Workflow

1. Clone repo
2. Create branch
3. Make changes
4. Commit
5. Push
6. Pull Request

7. Review & Merge

Explain *Why* Branching Matters

- Prevents breaking production
- Enables reviews
- Supports teamwork

✅ **Checkpoint Question**

Why **shouldn't** you commit directly to `main`?

🧪 Lab 2 — Clone Repo & Make Changes Locally (30 min)

Objective

Practice local Git usage.

Lab Steps

git clone https://github.com/tndagc-git/intern-training.git

cd intern-training

1. Create a new branch:

git checkout -b feature/add-notes

2. Edit README.md
3. Save changes
4. Commit:

git status

git add .
or
git add -A

git commit -m "Added intern notes"

5. Push branch:

git push origin feature/add-notes

✅ **Success Criteria**

- Branch exists in GitHub
- Commit message is meaningful

🧪 Lab 3 — Pull Requests & Code Review (20 min)

Objective

Learn collaboration and approvals.

Lab Steps

1. Open Pull Request in GitHub
2. Add description:
   - What changed?
   - Why?
3. Assign reviewer
4. Reviewer leaves comment
5. Merge PR

✅ **Success Criteria**

- PR created and merged
- Intern understands review process

📌 Section 3 — GitHub for IT & Cloud (15 min)

Show Real Examples

- Terraform repo
- Intune PowerShell scripts
- Documentation repo

Best Practices

✅ Clear commit messages
✅ Small changes
✅ Never store secrets

✅ Use `.gitignore`
✅ Use README files

Explain Secrets Handling

- Why secrets should **never** be in Git
- Use:
  - GitHub Secrets
  - Azure Key Vault
  - AWS Secrets Manager

✅ **Checkpoint Question**

What happens if you accidentally commit a password?

📝 Mini Quiz (10 min)

1. What is the purpose of a branch?
A. Delete files
B. Isolate changes ✅
C. Encrypt code
D. Publish apps

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2. True or False:
Commits should be large and infrequent
❌ False

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3. What command shows file status?
✅ `git status`

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4. Where should secrets be stored?
A. README
B. Code comments
C. GitHub Secrets ✅
D. Commit messages

🧪 Mini Assessment (30 min)

Scenario

“You are asked to add a troubleshooting note to a shared IT documentation repo.”

Intern must:

1. Create branch
2. Edit README
3. Commit changes
4. Open PR
5. Merge after review

✅ Pass Criteria

- No direct commits to main
- Clear commit message
- PR used correctly

📄 Optional Add-Ons (If Time Allows)

GitHub Features Overview

- Issues
- Projects
- Wiki
- GitHub Actions (high-level only)

GitHub + Azure or AWS Tie-In

- GitHub Actions → Azure or AWS
- Repo-based infrastructure
- CI/CD awareness